

Completed All tasks

1.below are the screen shots of the tasks results

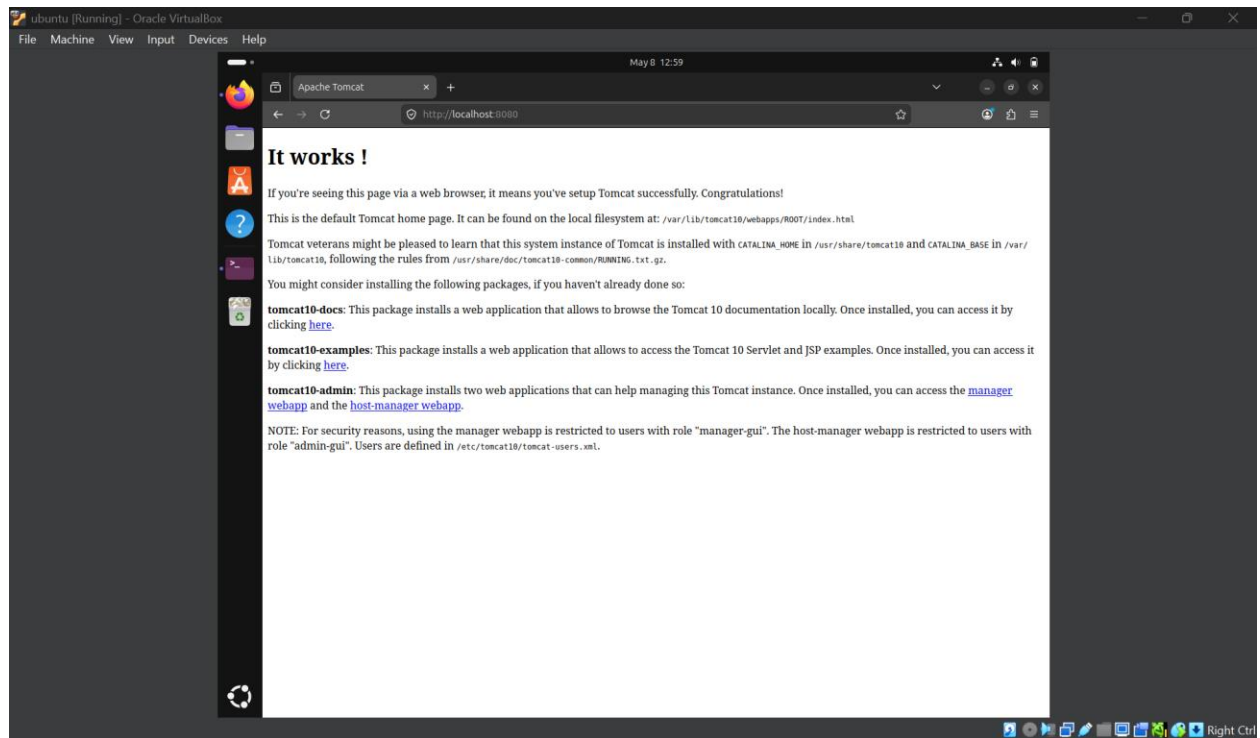
The first screenshot shows the terminal output for the first set of tasks. The user 'Techie' is logged in as 'vboxuser' on an Ubuntu machine. The terminal shows the following commands and their outputs:

```
Techie@ubuntu:~/home$ ls
> ls<
Techie@ubuntu:~/home$ ls
Techie@ubuntu:~/home$ cd
Techie@ubuntu:~/home$ ls
desktop Documents Downloads Music Pictures Public Templates Videos ss snap
Techie@ubuntu:~/home$ cd dassi
Techie@ubuntu:~/dassi$ ls
dassi dassi.tar s1 s1.tar
Techie@ubuntu:~/dassi$ cat s1
w1
e1
ed3
r4
t5
y6
u7
u7l8
lk8
ib1
800

de
efrr
gtrg
thtyh
yy
jyuj
vj
yuju
j
Techie@ubuntu:~/dassi$ awk '{print$2}'
```

The second screenshot shows the terminal output for the second set of tasks. The user 'Techie' is logged in as 'vboxuser' on an Ubuntu machine. The terminal shows the following commands and their outputs:

```
48 ls
49 cd
50 cp d1 /dassi
51 cd dassi
52 ls
53 cd dassi
54 ls
55 cd d1\
56 cd d1
57 ls
58 mkdir d3
59 ls
60 find -name s1
61 find / -name s1
62 cd
63 ls
64 cp s1 dassi
65 ls
66 mv s1 dassi
67 ls
68 cd dassi
69 ls
70 cd
71 touch s1.word
72 touch s1.word
73 ls
74 rm s1.word
75 ls
76 ll
77 chmod 722 dassi
78 ll
79 chmod 777 dassi
80 ll
81 date
82 uptime
83 ps -ef
84 ls
85 systemctl start tomcat10
86 apt install nginx
87 sudo apt install nginx
```

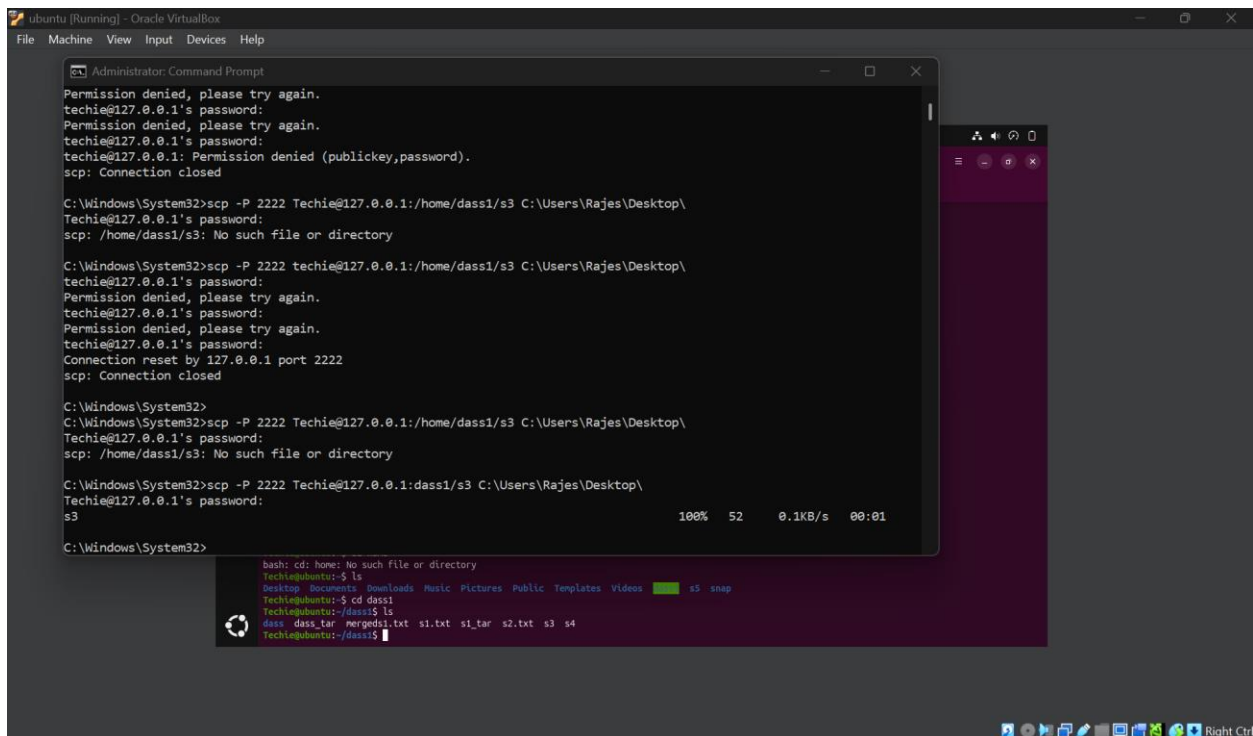
Successfully copied file from ubuntu vm to windows through command prompt using ssh

In vm i have change to ssh and added extra port 22 and local host 2222

And used command

```
scp -P 2222 Techie@127.0.0.1:dass1/s3 C:\Users\Rajes\Desktop\
```

Here scp i have used for copying and 2222 is local port number and next is user and vm ip address and files location then the windows path where i want to copy the file



The screenshot shows a Windows Command Prompt window titled 'Administrator: Command Prompt' and a terminal window titled 'ubuntu [Running] - Oracle VirtualBox'. The Command Prompt shows the execution of the scp command to copy a file from the Ubuntu VM to the Windows host. The terminal window shows the output of the scp command, indicating that the file 's3' has been successfully copied to the Windows host.

```
Permission denied, please try again.
techie@127.0.0.1's password:
Permission denied, please try again.
techie@127.0.0.1's password:
techie@127.0.0.1: Permission denied (publickey,password).
scp: Connection closed

C:\Windows\System32>scp -P 2222 Techie@127.0.0.1:/home/dass1/s3 C:\Users\Rajes\Desktop\
Techie@127.0.0.1's password:
scp: /home/dass1/s3: No such file or directory

C:\Windows\System32>scp -P 2222 Techie@127.0.0.1:/home/dass1/s3 C:\Users\Rajes\Desktop\
techie@127.0.0.1's password:
Permission denied, please try again.
techie@127.0.0.1's password:
Permission denied, please try again.
techie@127.0.0.1's password:
Connection reset by 127.0.0.1 port 2222
scp: Connection closed

C:\Windows\System32>
C:\Windows\System32>scp -P 2222 Techie@127.0.0.1:/home/dass1/s3 C:\Users\Rajes\Desktop\
Techie@127.0.0.1's password:
scp: /home/dass1/s3: No such file or directory

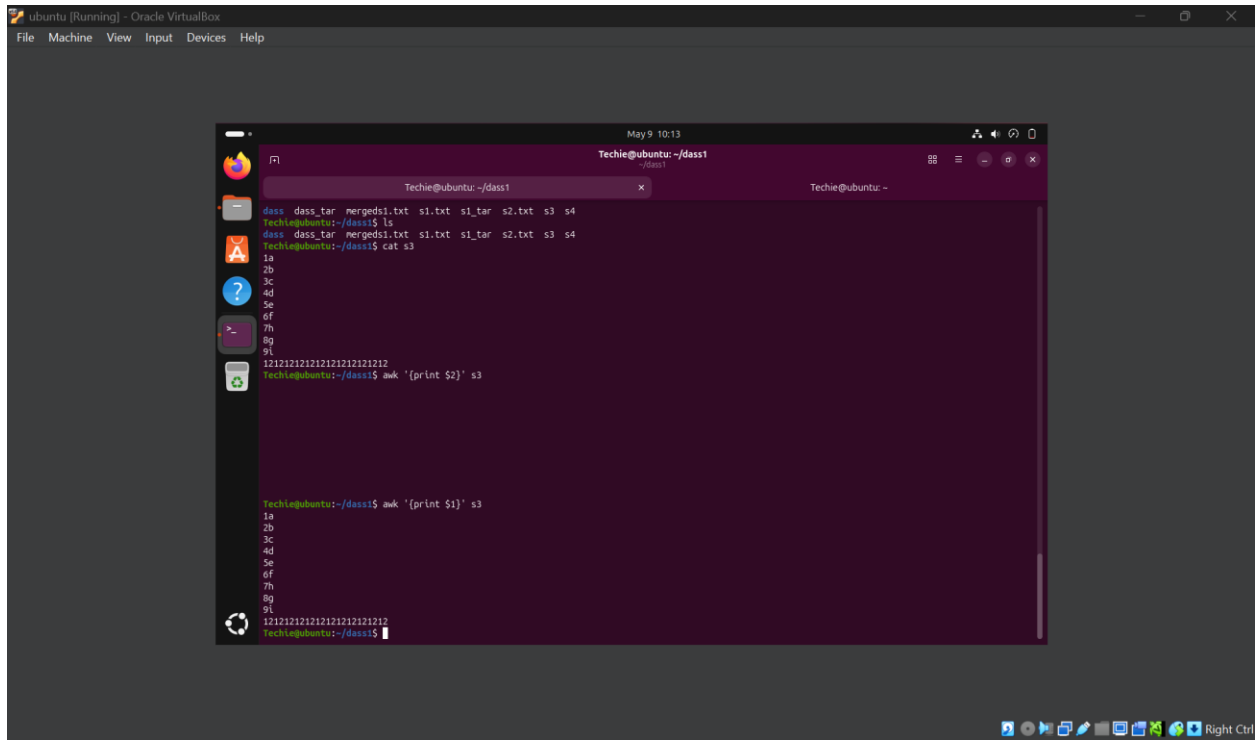
C:\Windows\System32>scp -P 2222 Techie@127.0.0.1:dass1/s3 C:\Users\Rajes\Desktop\
Techie@127.0.0.1's password:
s3                                     100% 52    0.1KB/s   00:01

C:\Windows\System32>
```

```
bash: cd: home: No such file or directory
Techie@ubuntu:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  s3  snap
Techie@ubuntu:~$ cd dass1
Techie@ubuntu:~/dass1$ ls
dass_dass_tar  mergeds1.txt  s1.txt  s1_tar  s2.txt  s3  s4
Techie@ubuntu:~/dass1$
```

Printed the particular columns using awk command

Awk '{ print \$linenumber}' file name



```
ubuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

May 9 10:13
Techie@ubuntu: ~/dass1
Techie@ubuntu:~/dass1
dass dass_tar merged1.txt s1.txt s1_tar s2.txt s3 s4
Techie@ubuntu:~/dass1$ ls
dass dass_tar merged1.txt s1.txt s1_tar s2.txt s3 s4
Techie@ubuntu:~/dass1$ cat s3
1a
2b
3c
4d
5e
6f
7h
8g
9l
1212121212121212121212121212
Techie@ubuntu:~/dass1$ awk '{print $2}' s3
1a
2b
3c
4d
5e
6f
7h
8g
9l
1212121212121212121212121212
Techie@ubuntu:~/dass1$ awk '{print $1}' s3
1a
2b
3c
4d
5e
6f
7h
8g
9l
1212121212121212121212121212
Techie@ubuntu:~/dass1$
```

Installed tomcat10 and edited the service file with java path

Grep -n(line num) filename

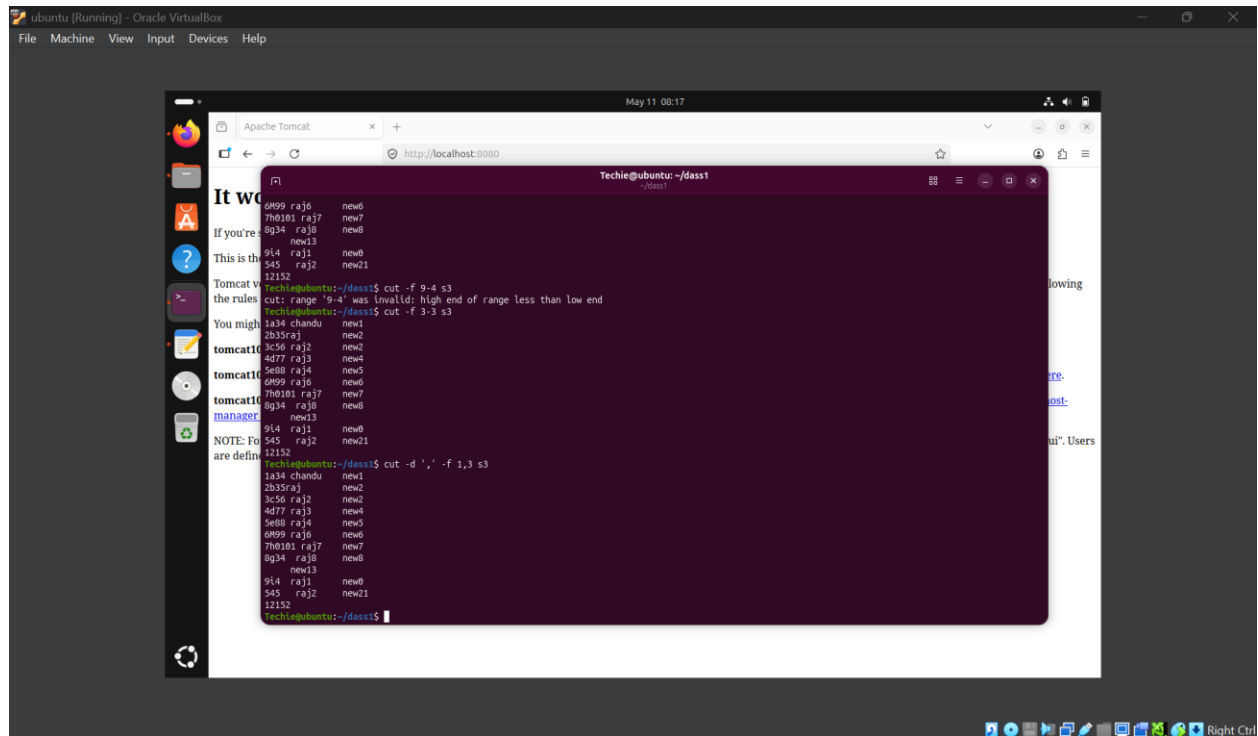
Printing specific lines from delimited files

We can use multiple commands

For example

Awk ,cut,

I have used cut command



The screenshot shows a terminal window titled 'Techie@ubuntu: ~/dass1' with the following commands and output:

```
cut -f 9-4 s3
cut: range '9-4' was invalid: high end of range less than low end
cut -f 3-3 s3
1a34 chandu new1
2b35raj new2
3c56 raj2 new2
4d77 raj3 new4
5d88 raj4 new3
6M99 raj6 new6
7h0101 raj7 new7
8g34 raj8 new8
9l4 raj1 new9
s45 raj2 new21
12152
```

The terminal window is overlaid on a web browser showing a page with text like 'It wo', 'If you're', 'This is the', 'Tomcat v', 'the rules', 'You migh', 'tomcat10', 'manager', and 'NOTE: Fo'. The browser's address bar shows 'http://localhost:8080'.

For the sum and avg i have used datamash if the file has only numbers

Command

Datamash sum (c.n) < filename

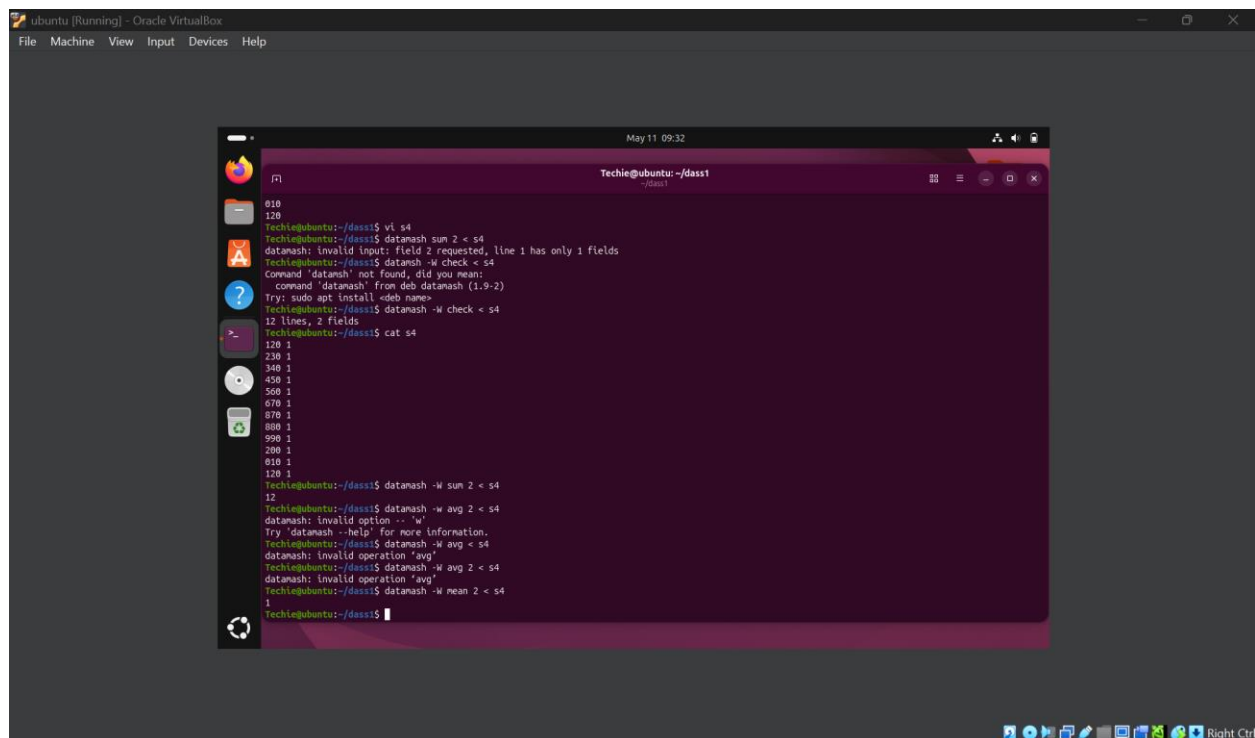
For average

Datamashf mean (c.n) < filename

We can also use

Datamash -W sum < filename

Herer -W acts as ignores spaces in text file



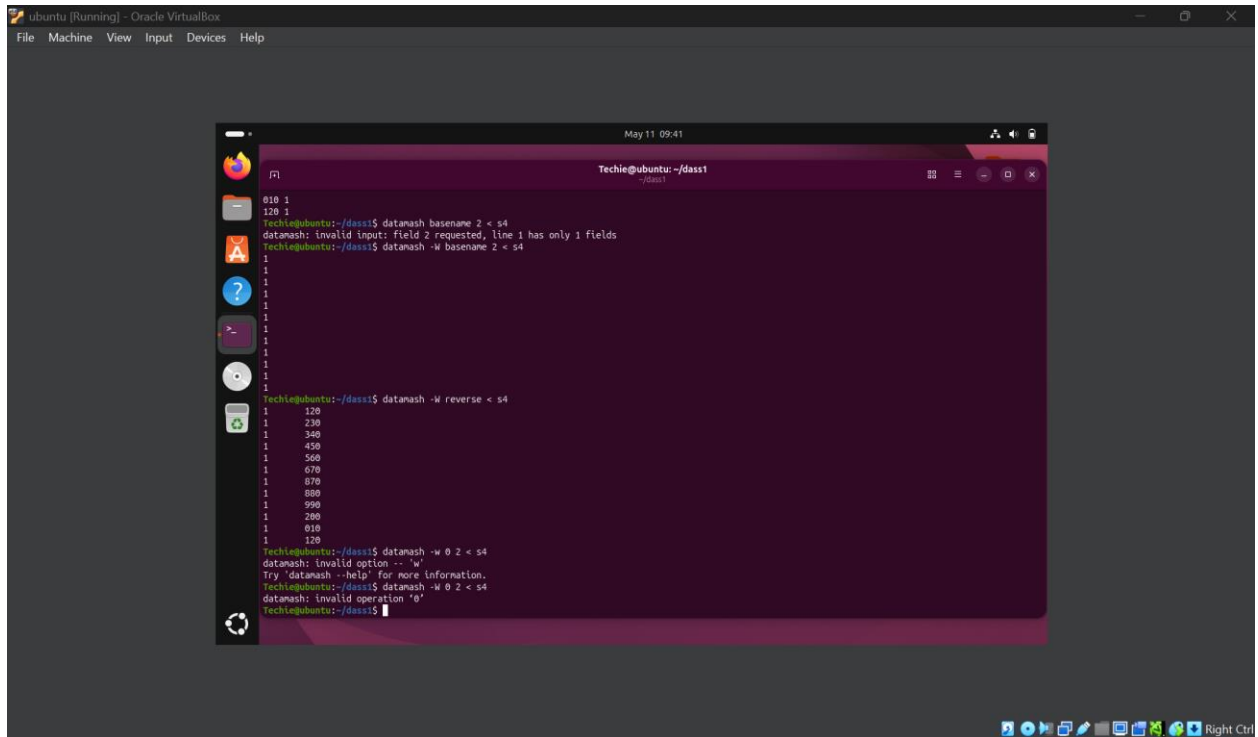
```
ubuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Techie@ubuntu: ~/datash
010
120
Techie@ubuntu:~/datash$ vi s4
datamash: invalid input: field 2 requested, line 1 has only 1 fields
Techie@ubuntu:~/datash$ datamash -W check < s4
Command 'datamash' not found, did you mean:
  command 'datamash' from deb datamash (1.9-2)
Try: sudo apt install <deb name>
Techie@ubuntu:~/datash$ datamash -W check < s4
12 lines, 2 fields
Techie@ubuntu:~/datash$ cat s4
120 1
230 1
340 1
450 1
560 1
670 1
780 1
890 1
990 1
200 1
010 1
120 1
Techie@ubuntu:~/datash$ datamash -W sum 2 < s4
12
Techie@ubuntu:~/datash$ datamash -W avg 2 < s4
datamash: invalid option -- 'w'
Try 'datamash --help' for more information.
Techie@ubuntu:~/datash$ datamash -W avg < s4
datamash: invalid operation 'avg'
Techie@ubuntu:~/datash$ datamash -W avg 2 < s4
datamash: invalid operation 'avg'
Techie@ubuntu:~/datash$ datamash -W mean 2 < s4
1
Techie@ubuntu:~/datash$
```

I have used datmash for reversing columns and pulling out specific name basename from columns

Command

Datamash -W reverse (c.n) < filename



To search and replace i have used sed command

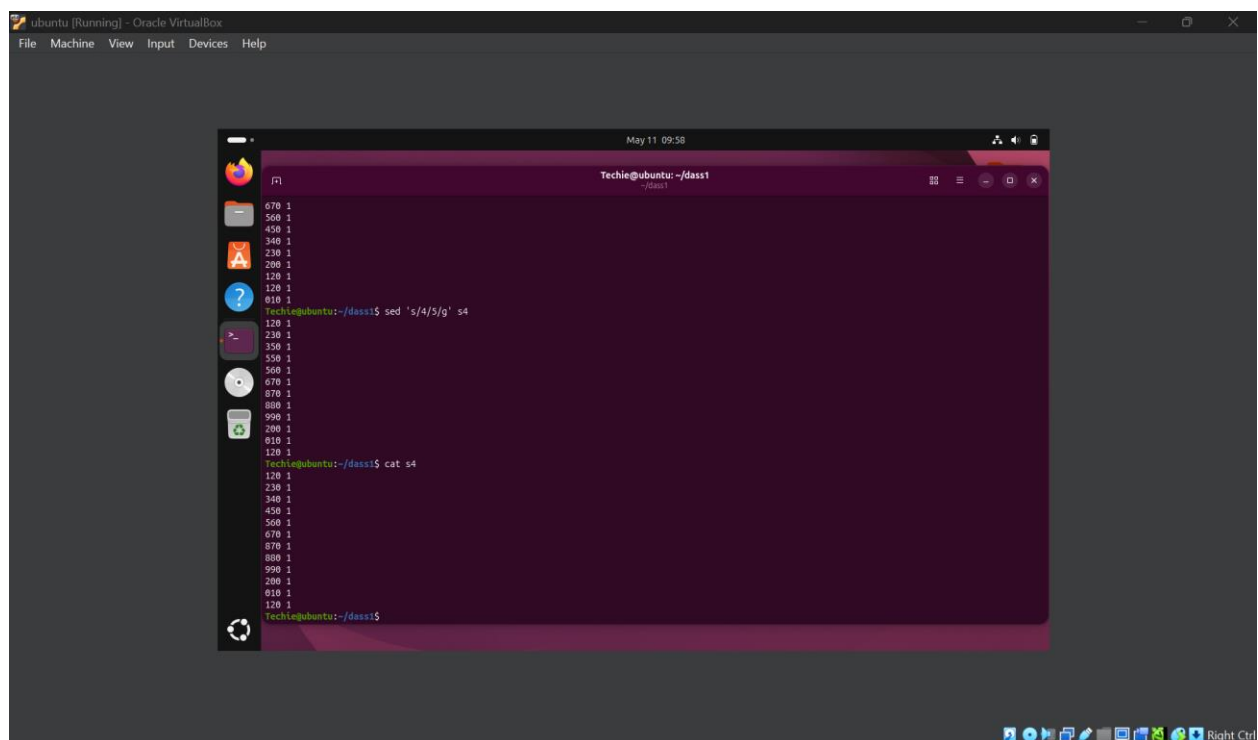
It will change the data just to show but actually it wont change

Command:

Sed 's/oldtext/newtext/g' filename

If i want to change the data permanently then i will use
"-i"

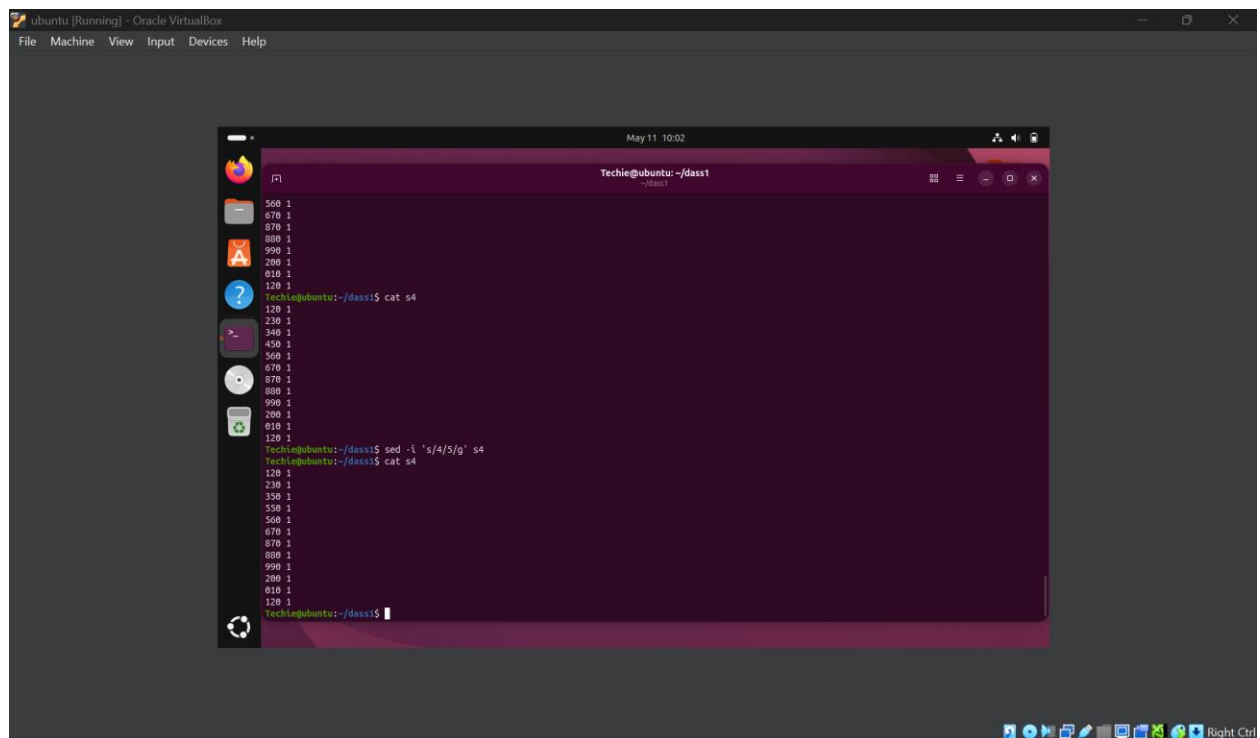
Sed -i 's/oldtext/newtext/g' filename



The screenshot shows a terminal window titled 'Techie@ubuntu: ~/dass1' with a dark purple background. The terminal displays the following commands and output:

```
128 1
508 1
458 1
348 1
238 1
208 1
128 1
128 1
818 1
Techie@ubuntu:~/dass1$ sed 's/4/5/g' s4
128 1
238 1
358 1
558 1
508 1
678 1
878 1
888 1
998 1
208 1
818 1
128 1
Techie@ubuntu:~/dass1$ cat s4
128 1
238 1
348 1
458 1
508 1
678 1
878 1
888 1
998 1
208 1
818 1
128 1
Techie@ubuntu:~/dass1$
```

The terminal window is running on an Ubuntu machine within an Oracle VM VirtualBox. The desktop environment includes a sidebar with application icons (Firefox, Files, LibreOffice, etc.) and a top status bar showing the date and time as 'May 11 09:58'.



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

May 11 10:02
Techie@ubuntu:~/dass1
508 1
678 1
878 1
888 1
998 1
208 1
018 1
128 1
Techie@ubuntu:~/dass1$ cat s4
128 1
238 1
348 1
458 1
568 1
678 1
878 1
888 1
998 1
208 1
018 1
128 1
Techie@ubuntu:~/dass1$ sed -i 's/4/5/g' s4
Techie@ubuntu:~/dass1$ cat s4
128 1
238 1
358 1
558 1
568 1
678 1
878 1
888 1
998 1
208 1
018 1
128 1
Techie@ubuntu:~/dass1$
```

Here i have used –i then data has been change permanently

To delete specific lines from text file

I will use sed command

Ex

Sed -i '13,20d' filename

Here it will delete 13 to 20 lines from the file

To delete the line except 13-20 i will use !

Ex

Sed -i '13-20!d' filename

Here it will delete lines every line except 13-20

And i want to just see the lines changes but not permanently delete i will not -i

The changes will be temporary not permanent

If i use -i then it will be permanent

```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

May 11 10:16
Techie@ubuntu: ~/dass1
gvfsvbdf
agvsadgv
asddgaegf
sdvsdv
12
23
Techie@ubuntu:~/dass1$ cat s4
120 1
230 1
350 1
550 1
560 1
670 1
670 1
670 1
800 1
990 1
200 1
010 1
120
defdsdv
erbfvdfcv
gvfsvbdf
agvsadgv
asddgaegf
sdvsdv
12
23
23
23
13
13
13
13
13
Techie@ubuntu:~/dass1$ sed
```

```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

May 11 10:17
Techie@ubuntu: ~/dass1
--posix specify the desired line-wrap length for the 'l' command
--posix disable all GNU extensions.
-E, -r, --regexp-extended use extended regular expressions in the script
(for portability use POSIX -E).
-s, --separate consider files as separate rather than as a single,
continuous long stream.
--sandbox operate in sandbox mode (disable e/r/w commands).
-u, --unbuffered load minimal amounts of data from the input files and flush
the output buffers more often
-z, --null-data separate lines by NUL characters
--help display this help and exit
--version output version information and exit
If no -e, -expression, -f, or --file option is given, then the first
non-option argument is taken as the sed script to interpret. All
remaining arguments are names of input files; if no input files are
specified, then the standard input is read.
GNU sed home page: <https://www.gnu.org/software/sed/>.
General help using GNU software: <https://www.gnu.org/gethelp/>.
Techie@ubuntu:~/dass1$ sed '13,201d' s4
defdsdv
erbfvdfcv
gvfsvbdf
agvsadgv
asddgaegf
sdvsdv
12
23
Techie@ubuntu:~/dass1$ cat s4
```

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

May 11 10:18

Techie@ubuntu: ~/dass1

```
010 1
120 1
Techie@ubuntu:~/dass1$ vi s4
Techie@ubuntu:~/dass1$ cat s4
120 1
230 1
350 1
550 1
500 1
670 1
870 1
800 1
990 1
200 1
010 1
120
defdsqv
efwsgrg
erbdrydfcv
grfsvdof
agvsadqv
asddgaegf
sdvsvd
12
23
23
23
13
13
13
13
13
Techie@ubuntu:~/dass1$ sed 14,20d s4
120 1
230 1
---
```

Right Ctrl

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

May 11 10:18

Techie@ubuntu: ~/dass1

```
Techie@ubuntu:~/dass1$ sed 14,20d s4
120 1
230 1
350 1
550 1
500 1
670 1
870 1
800 1
990 1
200 1
010 1
120
defdsqv
23
23
23
13
13
13
13
13
Techie@ubuntu:~/dass1$ sed 13,20d s4
120 1
230 1
350 1
550 1
500 1
670 1
870 1
800 1
990 1
200 1
010 1
120
23
```


Inserting text before or after pattern or particular line i will use sed command

Ex

Sed -i '/pattern/a new text' filename

A===after

I===before

--Here with this i will insert new text for after every pattern

Sed -i '/pattern/i newtext' filename

--here i will insert text before every new pattern with i

Sed -i 'linenumberfollowedbya newtext' filename

---here i will insert new line after a particular mentioned line

Sed -i 'linenumberi newtext' filename\

---here i will insert new text before a particular mentioned line

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

May 11 10:34

Techie@ubuntu: ~/dass1

```
13
13
13
13
Techie@ubuntu:~/dass1$ sed -i '/z/l new repeated pattern' s4
Techie@ubuntu:~/dass1$ cat s4
new repeated pattern
120 1
new repeated pattern
230 1
350 1
550 1
560 1
670 1
870 1
new pattern text
880 1
new pattern text
990 1
new repeated pattern
200 1
010 1
new repeated pattern
120
new inserted text
new repeated pattern
23
new repeated pattern
23
new repeated pattern
23
13
13
13
13
Techie@ubuntu:~/dass1$
```

Right Ctrl

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

May 11 10:40

Techie@ubuntu: ~/dass1

```
13
13
13
Techie@ubuntu:~/dass1$ sed -i '0/a new pattern text' s4
sed: -e expression #1, char 2: unknown command: '/'
Techie@ubuntu:~/dass1$ sed -i '/a new pattern text' s4
Techie@ubuntu:~/dass1$ cat s4
120 1
230 1
350 1
550 1
560 1
670 1
870 1
new pattern text
880 1
new pattern text
990 1
200 1
010 1
120
new inserted text
23
23
23
13
13
13
13
Techie@ubuntu:~/dass1$ sed -i '/z/l new repeated pattern' s4
Techie@ubuntu:~/dass1$ cat s4
new repeated pattern
120 1
new repeated pattern
230 1
350 1
```

Right Ctrl

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

May 11 10:40

Techie@ubuntu: ~/dass1

```
200 1
010 1
120 1
23
23
23
13
13
13
13
Techie@ubuntu:~/dass1$ sed -i '13l new inserted text' s4
Techie@ubuntu:~/dass1$ cat s4
120 1
230 1
350 1
550 1
500 1
670 1
870 1
800 1
900 1
200 1
010 1
120
new inserted text
23
23
23
13
13
13
13
Techie@ubuntu:~/dass1$ sed -i '0/a new pattern text' s4
sed: -e expression #1, char 2: unknown command: '/'
Techie@ubuntu:~/dass1$ sed -i '/a new pattern text' s4
Techie@ubuntu:~/dass1$ cat s4
```

Right Ctrl

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

May 11 10:41

Techie@ubuntu: ~/dass1

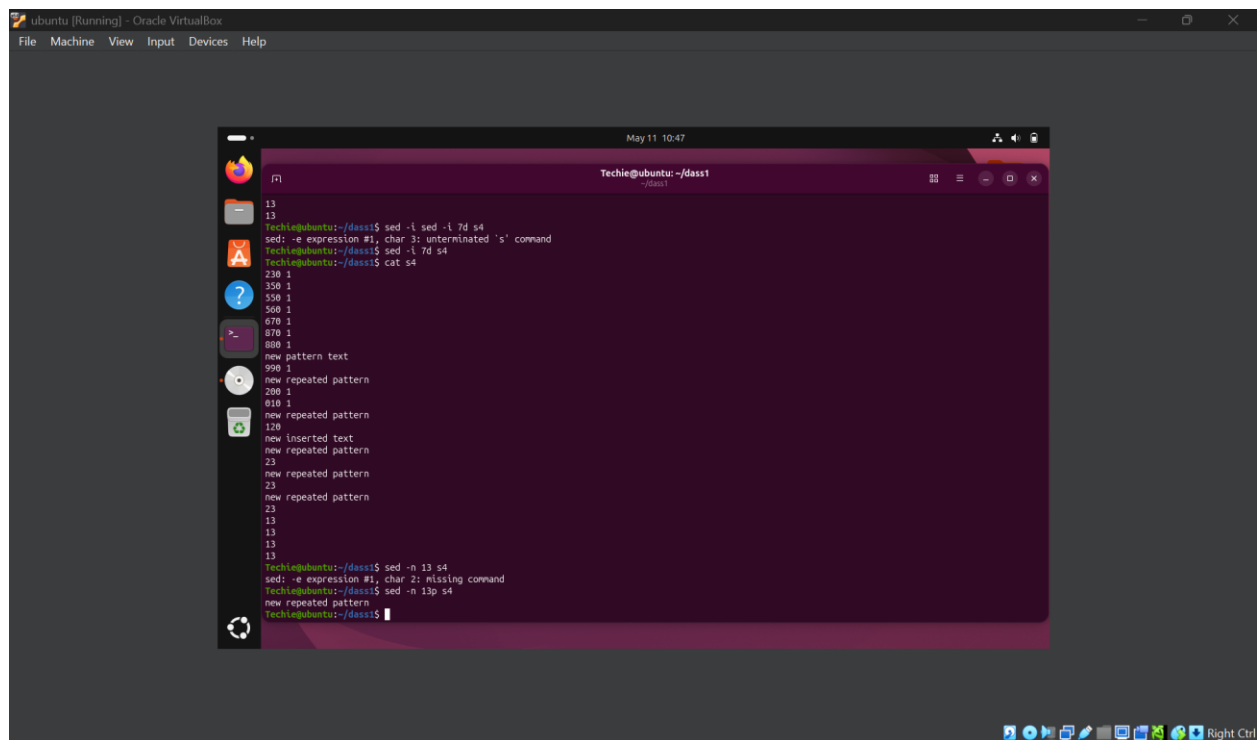
```
800 1
900 1
200 1
010 1
120
23
23
23
13
13
13
13
13
Techie@ubuntu:~/dass1$
Techie@ubuntu:~/dass1$ sed -i .bak 20d s4
Techie@ubuntu:~/dass1$ cat s4
120 1
230 1
350 1
550 1
500 1
670 1
870 1
800 1
900 1
200 1
010 1
120
23
23
23
13
13
13
13
Techie@ubuntu:~/dass1$ sed -i '13l new inserted text' s4
```

Right Ctrl

To print specific line from a text i have used sed command

Sed -n 'linenumfollowedby' filename

Ex sed -i 4p filename



```
ubuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

May 11 10:47
Techie@ubuntu: ~/dass1
13
13
Techie@ubuntu:~/dass1$ sed -i sed -i 7d s4
sed: -e expression #1, char 3: unterminated 's' command
Techie@ubuntu:~/dass1$ sed -i 7d s4
Techie@ubuntu:~/dass1$ cat s4
230 1
350 1
550 1
560 1
670 1
670 1
870 1
880 1
new pattern text
990 1
new repeated pattern
200 1
010 1
new repeated pattern
120
new inserted text
new repeated pattern
23
new repeated pattern
23
new repeated pattern
23
13
13
13
13
Techie@ubuntu:~/dass1$ sed -n 13 s4
sed: -e expression #1, char 2: missing command
Techie@ubuntu:~/dass1$ sed -n 13p s4
new repeated pattern
Techie@ubuntu:~/dass1$
```

5 use cases for sed

Sed -i linenumberfollowed by d filename

It will delete the particular line

Sed -i 'n,nd' filename

It will delete 1n to last n lines

Sed -i n!d filename

it will delete all lines except nline

Sed -i 'n,n!d' filename

it will delete all lines except the given lines

Sed -n np filename

It will print particualr line